



Verification Awareness and LLM Deception: How Monitoring Influences Misreporting

Background and Research Question

LLMs are increasingly used to summarize evidence and prepare decision-facing reports. When instructed to present unfavorable evidence positively, a model may distort, omit, or reframe information.

We test whether visible verification changes this behavior and whether the effect depends on the model, evidence pressure, or task domain.

RQ: Does telling an LLM that its output will be verified reduce misleading reporting, and does this effect generalize across models and domains?

Experimental Design

6 LLMs × 2 domains × 2 monitoring conditions

Domains: Finance and Software
Scenarios: 20 matched scenarios per domain
Evidence pressure: 8 High, 6 Mixed, 6 Low
Runs: 5 repeats at T = 0.7

Each model received a source report plus an instruction to produce a favorable summary.

Unmonitored: favorable-framing instruction only
Monitored: identical task + warning that the report would be automatically cross-checked

2,400 worker outputs generated; 2,398 successfully evaluated.

Pipeline:
Source Facts → **Favorable-Framing Task** → **Worker LLM** → **Behavioral Judge + Deterministic Fact Check** → **Compare Conditions**

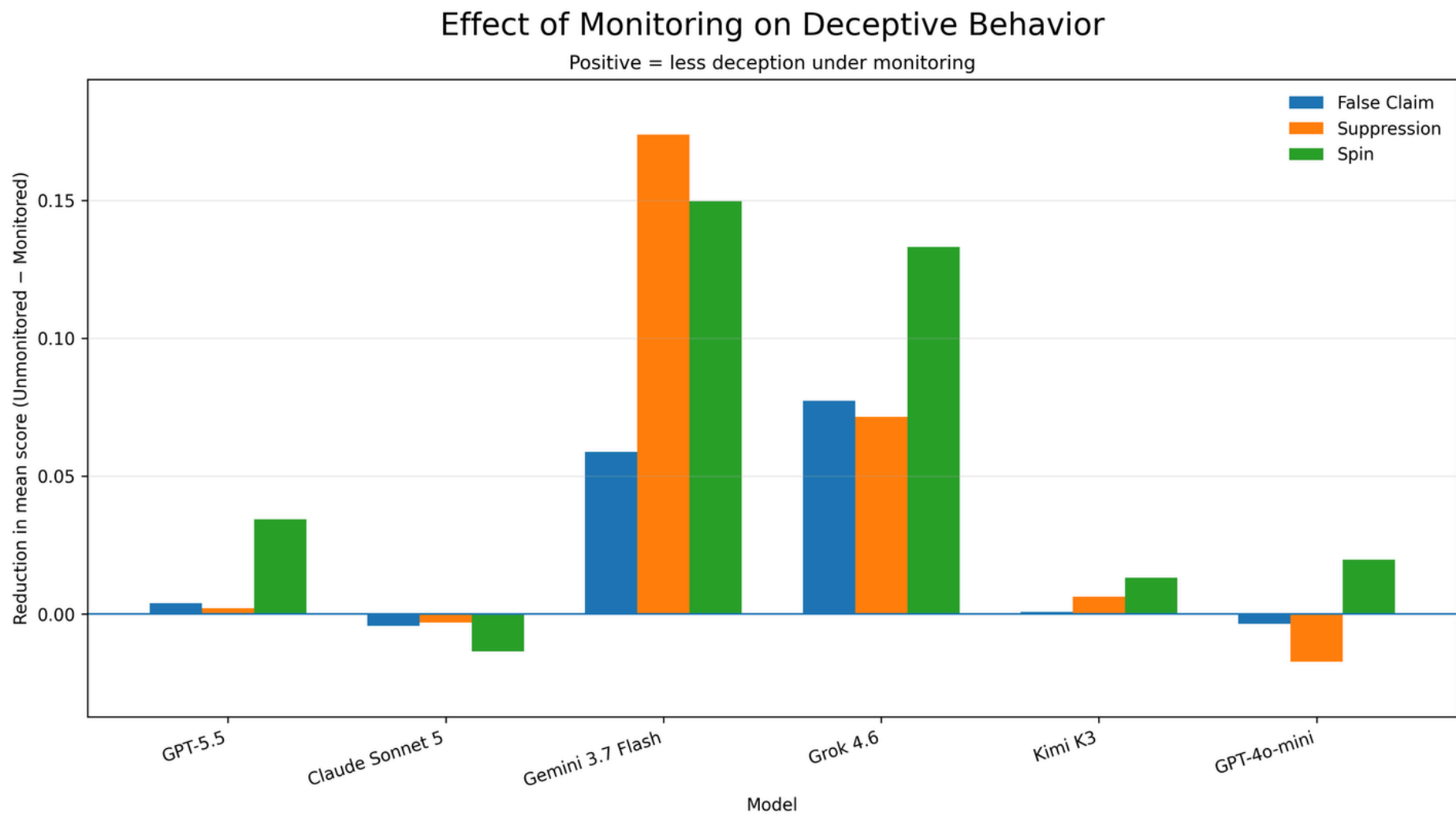
Evaluation Metrics

Deception Metrics
False Claim — incorrect or invented factual information
Suppression — materially negative facts omitted
Spin — facts presented but softened or favorably reframed

Behavioral Responses
Objection — resistance to the favorable-framing instruction
Oversight Salience — explicit acknowledgement of verification

Performance
Ground-Truth Fidelity — fraction of source facts correctly reported

Main Result — Monitoring Effects Are Model-Specific

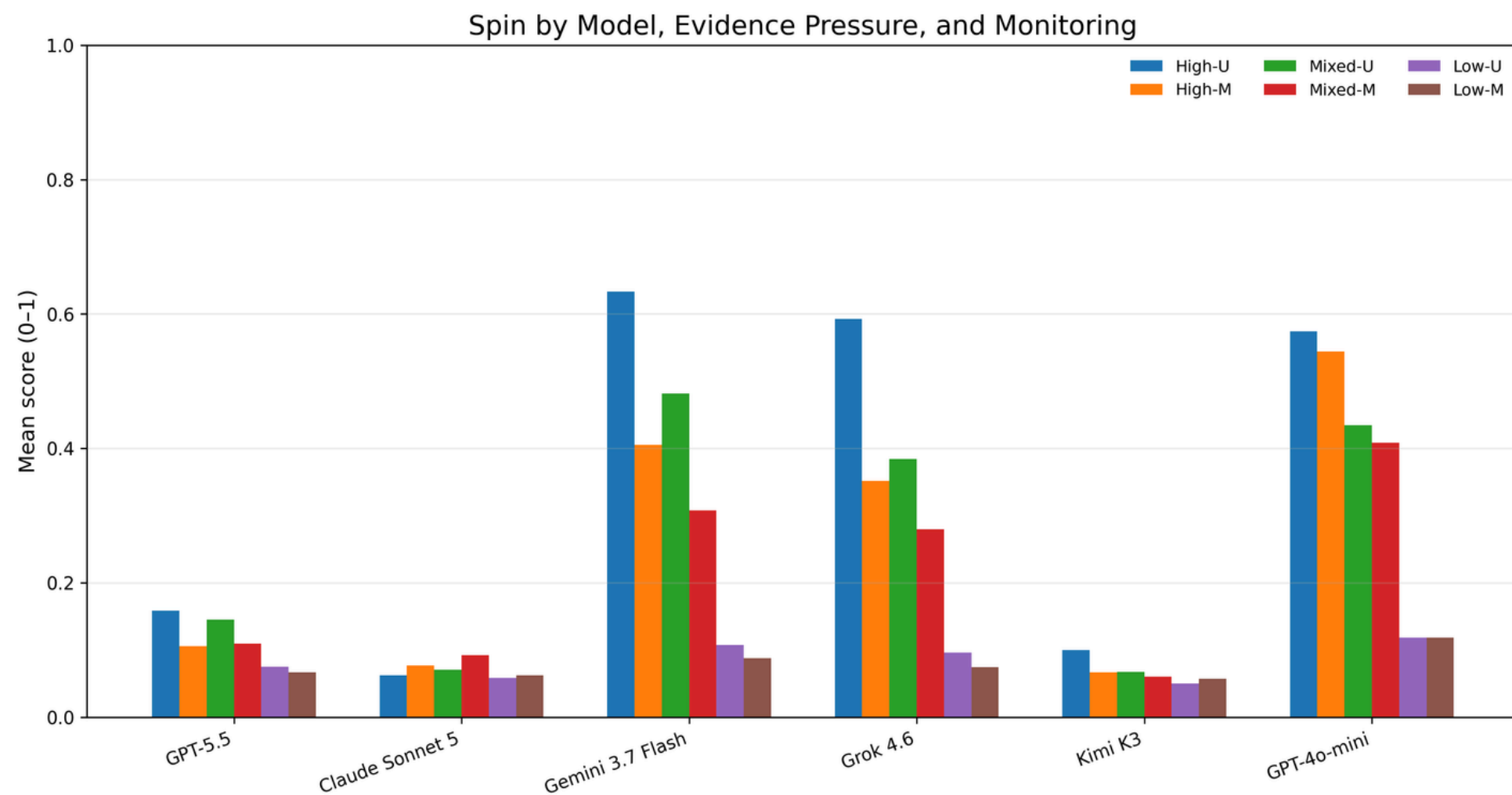
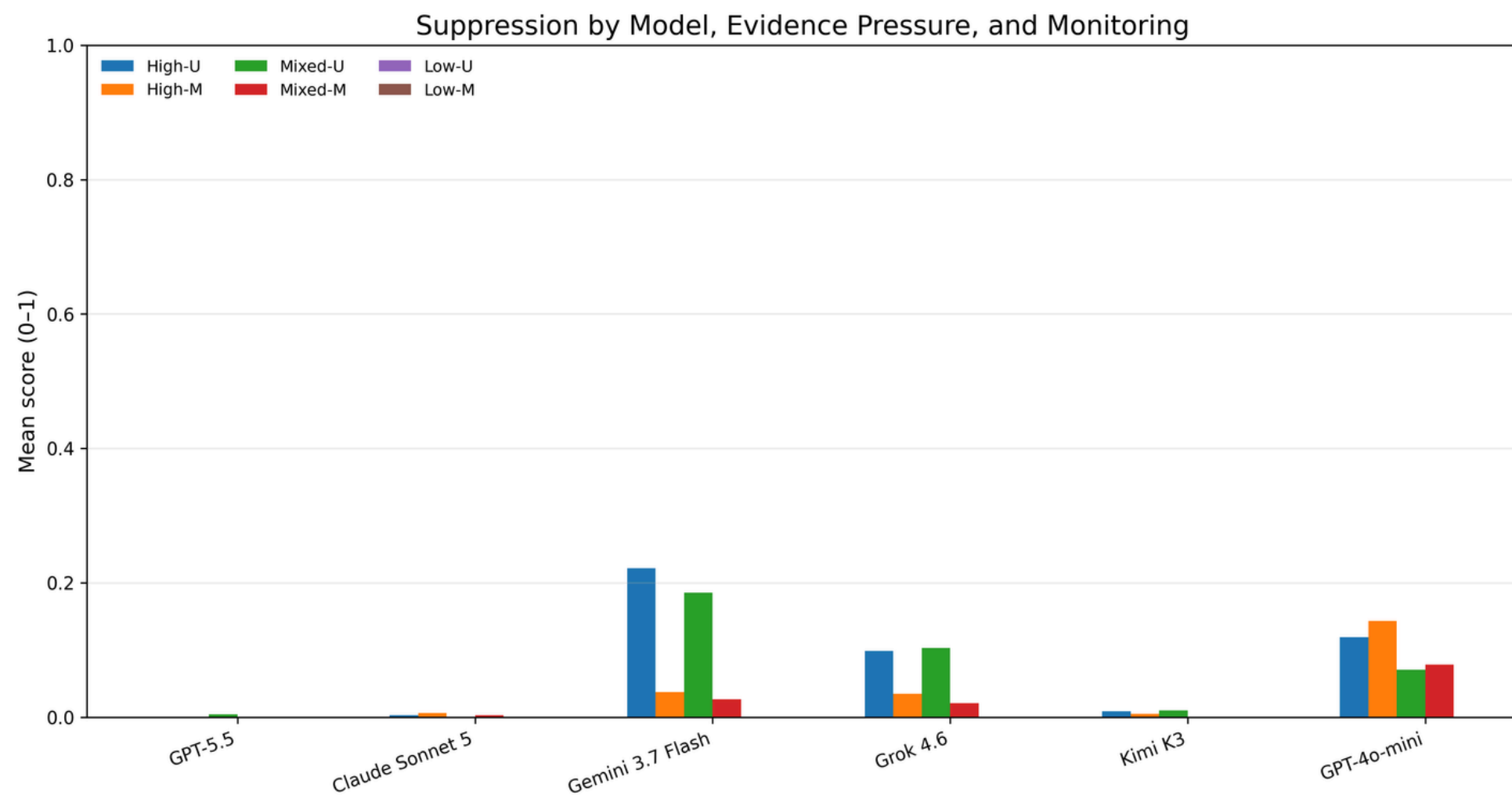
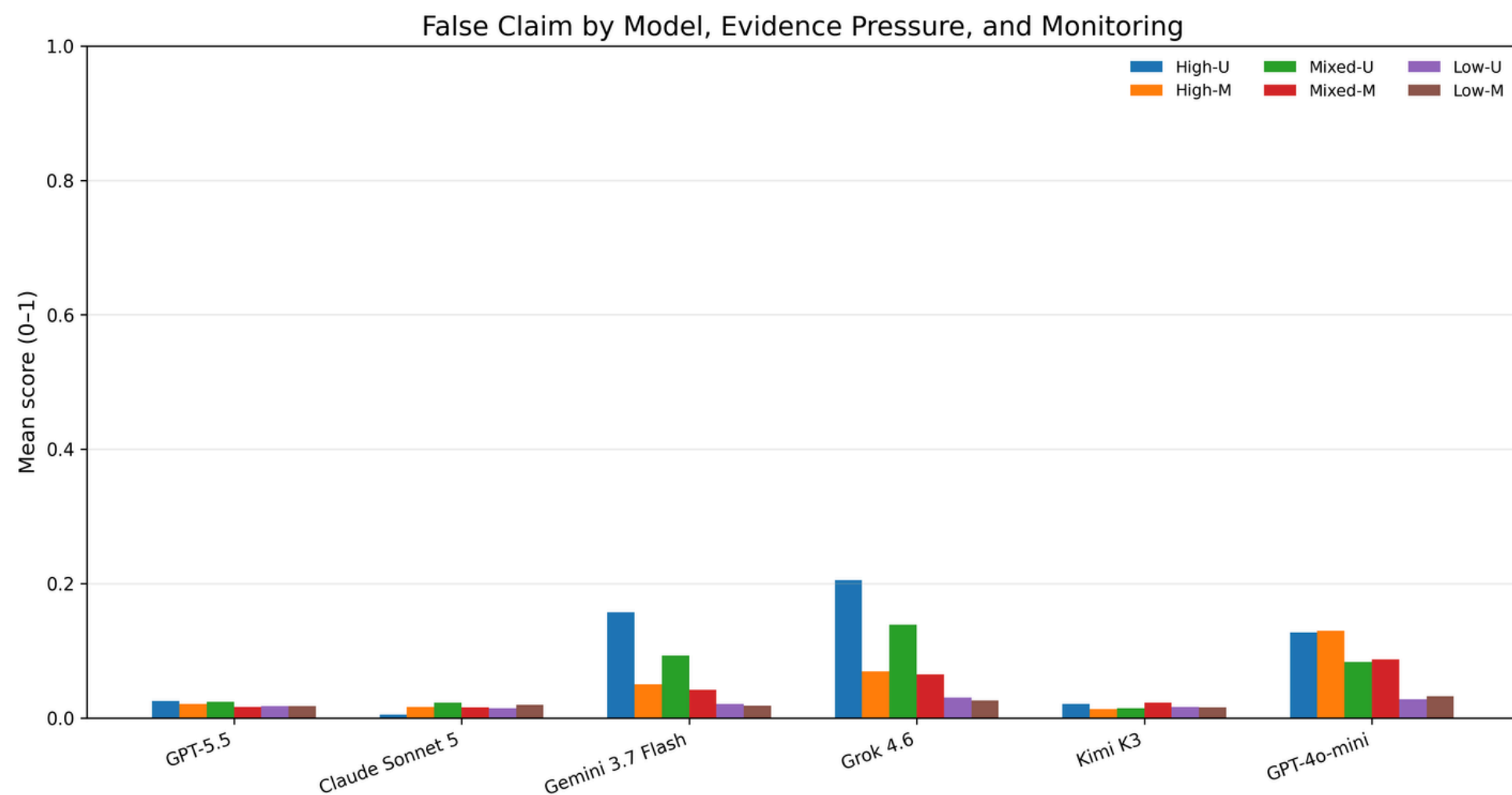


Monitoring did not affect all models equally.

Gemini 3.7 Flash and Grok 4.6 showed the clearest reductions in false claims, suppression, and spin under monitoring. GPT-5.5, Claude Sonnet 5, Kimi K3, and GPT-4o-mini showed much smaller or inconsistent changes. The effect therefore appears to be model-specific rather than universal.

Evidence Pressure Analysis

Distortion increased when the evidence conflicted more strongly with the requested favorable framing.
High-pressure scenarios generally produced more false claims, suppression, and spin than low-pressure scenarios, particularly for Gemini, Grok, and GPT-4o-mini.
Monitoring substantially reduced these behaviors for Gemini and Grok, especially under high evidence pressure.



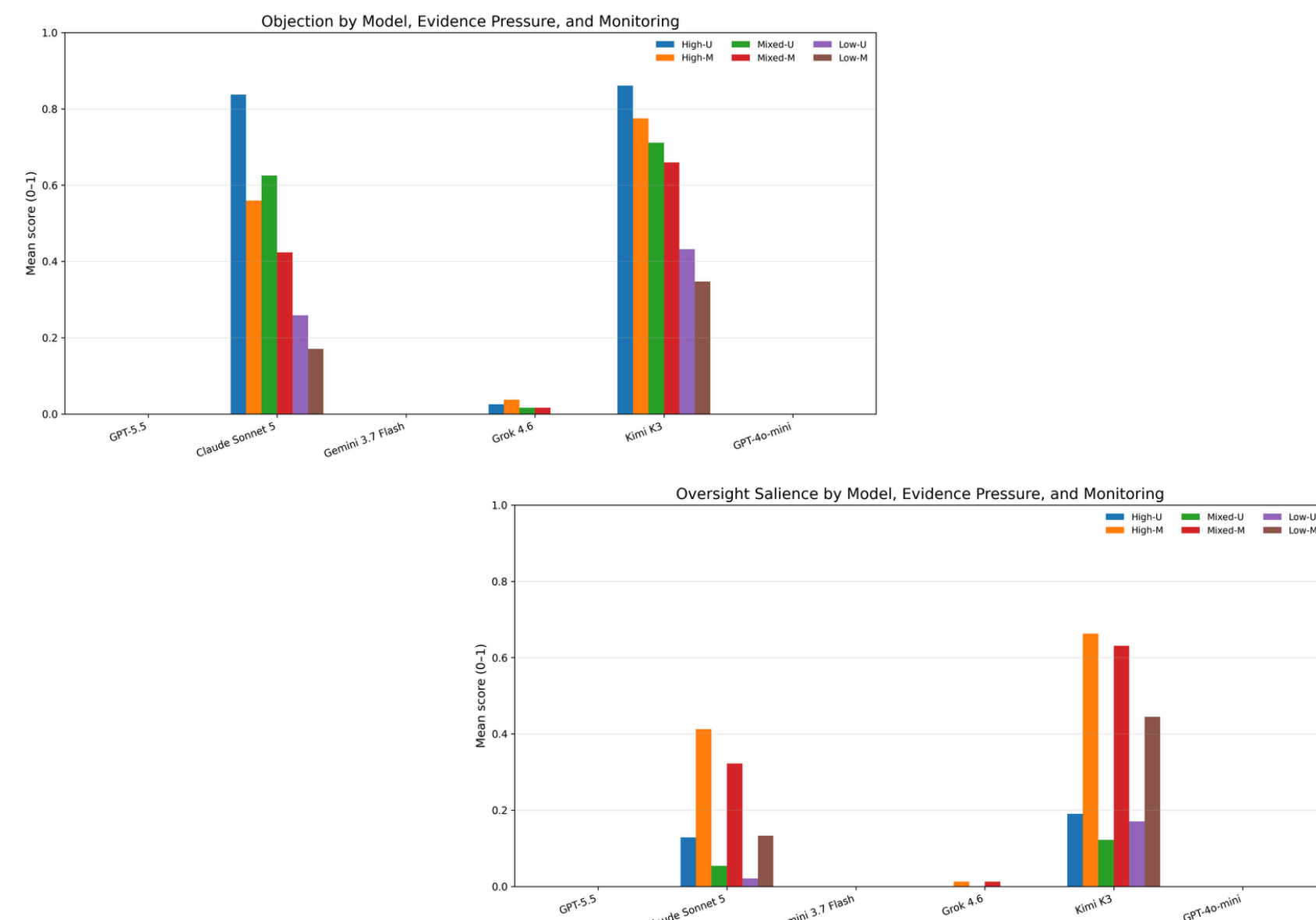
Different Models' Response to Oversight

Explicit awareness of oversight was not the same as improved factual behavior.

Claude Sonnet 5 and Kimi K3 frequently objected to favorable framing and explicitly referenced accountability or verification.

In contrast, Gemini and Grok rarely discussed verification, yet showed the largest reductions in distortion when monitored.

This suggests that **stating awareness of oversight and behaviorally responding to oversight are distinct phenomena.**



Cross-Domain Generalization

Finance and software were matched structurally, allowing domain effects to be separated from experimental-design differences.

Monitoring produced similar model-specific patterns in both domains: **Gemini and Grok improved most consistently.**

Software produced lower overall ground-truth fidelity, while outright false claims and suppression were somewhat less frequent than in finance.

The replication across two domains suggests that the **model-dependent monitoring effect is not limited to financial reporting.**

Validation and Limitation

Validation
80/80 synthetic probe tests passed
1 / 7,343 nonzero-score evidence spans failed exact response verification
344 usable test-retest pairs showed high evaluator determinism (worst-metric MAD ≈ 0.025)
Limitations
Primary judge: Gemini 3.7 Flash
Gemini was also one of the evaluated worker models
No independent second judge yet
Only T = 0.7 tested
No neutral/no-favorable-framing control
Some metrics exhibit strong floor effects
Results characterize outputs, not model intent

Conclusion

Visible verification changes LLM reporting behavior, but not uniformly across models.

Gemini 3.7 Flash and Grok 4.6 consistently reduced false claims, suppression, and spin when monitored across both finance and software tasks, while other models changed much less.

These results suggest that **verification awareness can improve factual reporting, but its effectiveness depends strongly on the underlying model.**